

# Variable Scope: Managing Data Access in Trading Systems

When building trading systems, data management is critical. You need to control which parts of your program can access which data. Variable scope—the rules determining where variables are accessible—is how Python manages this.

## Local Variables: Isolated Within Functions

Variables defined inside a function are local to that function—they can't be accessed from outside:

```
1. def analyze_volatility(prices):
2.     # This is a local variable - only exists inside this function
3.     daily_returns = []
4.
5.     for i in range(1, len(prices)):
6.         daily_return = (prices[i] - prices[i-1]) / prices[i-1]
7.         daily_returns.append(daily_return)
8.
9.     # Calculate volatility as standard deviation of returns
10.    mean_return = sum(daily_returns) / len(daily_returns)
11.    squared_deviations = [(r - mean_return) ** 2 for r in daily_returns]
12.    variance = sum(squared_deviations) / len(daily_returns)
13.    volatility = variance ** 0.5
14.
15.    return volatility
16.
17. # Call the function
18. stock_prices = [100, 102, 99, 101, 103, 102, 105]
19. vol = analyze_volatility(stock_prices)
20.
21. print(f"Volatility: {vol:.2%}")
22.
23. # This would cause an error - daily_returns is not accessible here
24. # print(daily_returns)
```